Week 6: Number theory

BSC 129 / BSC 136

- Divisibility, Division Algorithm
- Modular Arithmetic, Congruence
- Integer Expansion and Application: Fast exponentiation
- Primes and composites, GCD
- Euclidean Algorithm, Bezout Theorem
- Solving Congruences, Chinese Remainder Theorem

4.1 Divisibility

Definition

If a and b are integers with $a \neq 0$,

- a **divides** b means there is an integer k such that $b = ak$.

$$b / a = \frac{b}{a} = k$$

4.1 Divisibility

Definition

If a and b are integers with $a \neq 0$,

- a **divides** b means there is an integer k such that $b = ak$.
- a is a **factor** or **divisor** of b.
- b is a **multiple** of a.
- We denote this by $a \mid b$.
- We also say that b is **divisible** by a.

Easy facts

Let a, b and c be integers with $a \neq 0$. Then

- If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.

a l b means b = ak for some integer k, and
a l c means c = am for some integer m, then
b + c = ak + am = a(k+m), with k + m an integer, then means a l (b+c).

Easy facts

Let a, b and c be integers with $a \neq 0$. Then

- If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
- If $a \mid b$, then $a \mid bc$ for all integers c.

  a l b means b = ak for some integer k, and
  bc = (ak)c = a(kc) where kc is an integer, and so bc is a multiple of a.

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- If $a \mid b$, then $a \mid bc$ for all integers c.
- If $a \mid b$ and $b \mid c$, then $a \mid c$.

  a l b means b = ak for some integer k, and
  b l c means c = bm for some integer m, then c = bm = (ak)m = a(km)
  where km is an integer, and so c is a multiple of a.

Easy facts

Let a, b and c be integers with $a \neq 0$. Then

- If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
- If $a \mid b$, then $a \mid bc$ for all integers c.
- If $a \mid b$ and $b \mid c$, then $a \mid c$.
- If $a|b$ and $a|c$, then $a|(mb + nc)$ for all integers m and n.

linear combination of b and c

a l b means b = ak for some integer k, and
a l c means c = aj for some integer j, then
mb + nc = m(ak) + n(aj) = a (mk+nj), with mk+nj is an integer,
and so mb + nc is a multiple of a.

The Division Algorithm

Let a be an integer and d a positive integer, then there exist unique integers q and r, (d is not zero)

$$\text{with } 0 \leq r < d, \text{ such that } a = dq + r.$$

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- q is the called the **quotient**. q is the integer part of a/d, or we can write q = floor(a/d).
- r is called the **remainder**. r = a — dq.

When r is 0, then a = dq, it means a is a multiple of d.

a = 7, d= 3, then 7 = 3 x 2 + 1 (not 3 x 1 + 4 because the remainder is too large)

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Example. Find the quotient and remainder when 27 is divided by 8.

27/ 8 = 3.375 27 = 8 x 3 + 3, q= 3, r =3.

The Division Algorithm

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Example. Find the quotient and remainder when 27 is divided by 8.

Example. Find the quotient and remainder when -27 is divided by 8.

$-27 / 8 = -3.375$

$-27 = 8 \times (-3) - 3 = 8 \times (-4) + 5.$ $q = -4, r = 5.$

Example. How many positive integers less than 1000 are divisible by 7?

Example. Let n and d be positive integers. How many positive integers at most n are divisible by d? floor(n/d)